

ENTHUSIAST, LADER & ACHIEVER COURSE

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : DRILL TEST # 09

Test Pattern : NEET (UG)

TEST DATE : 19-04-2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	2	2	2	2	3	2	1	4	2	1	2	1	4	1	1	1	3	4	2	1	2	4	4	4	4	1	4	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	2	2	3	3	1	1	4	4	1	3	2	4	4	2	1	1	3	3	4	3	2	2	1	1	4	3	4	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	3	4	3	3	3	3	4	4	3	3	3	4	4	4	2	2	1	3	4	3	2	4	1	1	2	3	4	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	1	4	4	3	2	4	1	3	2	1	4	3	4	3	3	4	2	1	2	2	2	1	2	4	1	4	4	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	3	2	2	2	2	3	1	3	1	4	1	3	2	1	1	2	3	4	4	3	4	2	3	3	3	3	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	2	1	3	1	2	4	2	4	3	3	2	3	2	2	2	4	3	2	1	2	4	3	4	3	1	4	3	2	3

HINT - SHEET

1. Ans (2)

$$I = \frac{1}{2}MR^2$$

When the cylinder rolls down the inclined plane from a vertical height h its potential energy Mgh is partly converted into translational kinetic energy $\frac{1}{2}Mv^2$ and partly into rotational kinetic energy $\frac{1}{2}I\omega^2$. Therefore,

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2$$

$$= \frac{1}{2}M(R\omega)^2 + \frac{1}{2}MR^2\omega^2 \quad (\because R\omega = v)$$

$$Mgh = \frac{3}{4}MR^2\omega^2$$

$$\omega = \sqrt{\frac{4gh}{3R^2}} = \frac{2}{R}\sqrt{\frac{gh}{3}}$$

2. Ans (4)

$$r \propto A^{1/3}$$

3. Ans (2)

$$P = \left(\frac{1}{f_1} + \frac{1}{f_2}\right) \times 100 \text{ D if } f_1 \text{ \& } f_2 \text{ are cm}$$

$$\frac{1}{f_1} = \left(\frac{3}{2} - 1\right) \left(0 - \frac{1}{-10}\right) = \frac{1}{20}$$

$$\frac{1}{f_2} = (2 - 1) \left(-\frac{1}{10} - 0\right) = -\frac{1}{10}$$

$$D = \left(\frac{1}{20} - \frac{1}{10}\right) \times 100 = -5D$$

4. Ans (2)

By balancing equation.



$$Z = 92 - 56 = 36, A = 235 + 1 - 144 - 3 = 89$$

5. Ans (2)

$$E = \frac{V_0}{d} \Rightarrow V_0 = Ed$$

$$= (6 \times 10^5) \times (500 \times 10^{-9})$$

$$= 0.3 \text{ V}$$

6. Ans (2)

Use $E = I(R + r)$

$E = 0.5(2 + r) \dots\dots(1)$

$E = 0.25(5 + r) \dots\dots(2)$

by eq (1) & (2) $E = 1.5 \text{ V}$

7. Ans (3)

$\beta_1 = \frac{\lambda_1 D}{d} = \frac{6000 \times 10^{-10} \times 1}{10^{-3}} = 0.6 \text{ mm}$

$\beta_2 = \frac{\lambda_2 D}{d} = 0.45 \text{ mm}$

Let n_1 th maxima of λ_1 and n_2 th maxima of λ_2 coincide at a position y .

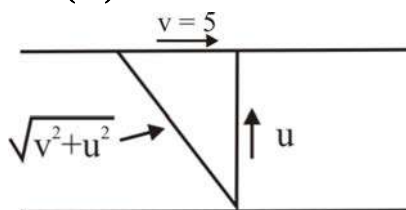
then, $y = n_1 \beta_1 = n_2 \beta_2 = \text{LCM of } \beta_1 \text{ and } \beta_2$

$\Rightarrow y = \text{LCM of } 0.6 \text{ mm and } 0.45 \text{ mm}$

$y = 1.8 \text{ mm}$

At this point 3rd maxima for $6000 \text{ \AA}$ & 4th maxima for $4500 \text{ \AA}$ coincide

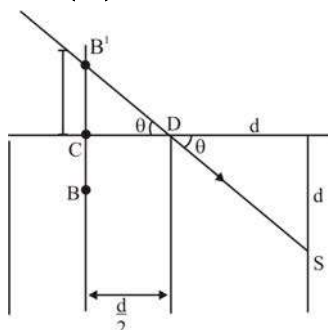
8. Ans (2)



$u = \frac{60}{5} = 12 \text{ m/s}, v = 5 \text{ m/s}$

$\sqrt{v^2 + u^2} = \sqrt{5^2 + 12^2} = 13 \text{ m/s}$

9. Ans (1)



$\tan \theta = \frac{d}{d/2} = 1 = \frac{CB'}{CD} = \frac{CB'}{d/2}$

$CB' = \frac{d}{2} \Rightarrow CB = \frac{d}{2} = 1 \text{ m}$

10. Ans (4)

$y = \overline{\overline{A + B}} = \overline{\overline{A}} + \overline{\overline{B}} = \overline{AB}$ NAND gate

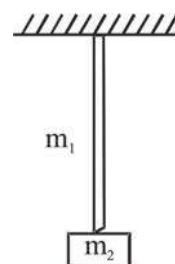
11. Ans (2)

$V = \sqrt{\frac{T}{\mu}} \Rightarrow V_1 = \sqrt{\frac{m_2 g}{\mu}}$

$\Rightarrow V_2 = \sqrt{\frac{(m_1 + m_2)g}{\mu}}$

$V = f\lambda$

$\lambda \propto V$



$\frac{\lambda_1}{\lambda_2} = \frac{V_1}{V_2}$

$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 + m_2}{m_2}}$

12. Ans (1)

$g = \frac{4}{3}\pi R \rho g$

$g \propto \rho$

13. Ans (2)

$\lambda_1 = 1 \text{ \AA}$

$\lambda_2 = 0.5 \text{ \AA}$

$\therefore V = \frac{150}{\lambda_e^2}$

$V_1 = \frac{150}{(1)^2} = -150 \text{ volt}, E_1 = 150 \text{ eV}$

$V_2 = \frac{150}{(0.5)^2} = \frac{150}{0.25} = 600 \text{ volt}, E_2 = 600 \text{ eV}$

$\Delta E = E_2 - E_1 = 600 - 150 = 450 \text{ eV}$

14. Ans (1)

Electric field lines are coming out from positive charge and ends to negative charge, so A is positive, B is negative total number of electric field lines are more A than B so $|A| > |B|$.

15. Ans (4)

$$g_E = \frac{GM_E}{R_E^2}, \quad g_M = \frac{GM_M}{R_M^2}$$

$$g_m = \frac{G(0.1M_E)}{(0.5R_E)^2} = 0.4g_E$$

16. Ans (1)

Work against friction is

$$W_f = fd = \mu Nd = \mu mgd \cos \theta (\because N = mg \cos \theta)$$

$$= 0.1 \times 1 \text{ kg} \times 10 \text{ ms}^{-2} \times \cos 30^\circ \times 10 \text{ m}$$

$$= 8.7 \text{ J}$$

17. Ans (1)

For floating body $W = Th = V_{in} \rho_L g$

$$Ah \rho_B g = (Ah) \times 3 \rho_B g$$

$$\frac{h}{H} = \frac{1}{3}$$

$$\text{Exposed height ratio} = \frac{H-h}{H} = \frac{3-1}{3} = \frac{2}{3}$$

18. Ans (1)

$$\int ds = \int_1^2 v dt = \int_1^2 (4t^3 + 3t^2 - 1) dt$$

$$s = (t^4 + t^3 - t)_1^2 = 21 \text{ m}$$

19. Ans (3)

$$C = \sqrt{\frac{3PV}{M}}$$

$$= \sqrt{\frac{3 \times 24 \times 10^5 \times 10 \times 100}{20}} \text{ cms}^{-1}$$

$$= 6 \times 10^4 \text{ cm s}^{-1} = 600 \text{ ms}^{-1}$$

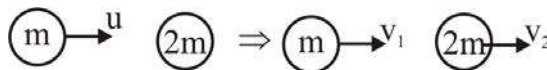
20. Ans (4)

$$a = \frac{5g - 4g}{9} = \frac{g}{9}$$

$$T_{AB} - 1g = 1 \times \frac{g}{9} \Rightarrow T_{AB} = \frac{10g}{9}$$

21. Ans (2)

Let mass of neutron = m, then mass of deuteron = 2m



By momentum conservation

$$mu = mv_1 + 2mv_2 \Rightarrow u = v_1 + 2v_2 \dots (1)$$

$$e = \frac{v_2 - v_1}{u} = 1 \text{ (Elastic collision)} \Rightarrow u = v_2 - v_1 \dots (2)$$

$$\text{by equation (1) \& (2)} \quad v_1 = -\frac{u}{3}$$

$$\text{Fractional loss of KE of neutron} = \frac{KE_i - KE_f}{KE_i}$$

$$= \frac{\frac{1}{2}mu^2 - \frac{1}{2}m\left(\frac{u}{3}\right)^2}{\frac{1}{2}mu^2} = 1 - \frac{1}{9} = \frac{8}{9}$$

22. Ans (1)

According to question

$$(\pi R^2) \frac{dB}{dt} = 2\pi rE$$

where $B = \mu_0 nI$

$$\text{Therefore } E = \frac{\mu_0 n I_0 \omega R^2}{2r} \sin \omega t$$

23. Ans (2)

The force acting on the elementary portion of the current carrying conductor is given as

$$dF = i(dr)B \sin 90^\circ$$

$$= iB dr$$

The torque applied by dF about O.

$$d\tau = r dF$$

The total torque about O,

$$\tau = \int d\tau = \int_0^1 r(iB)dr = \frac{iBL^2}{2}$$

The angular acceleration,

$$\alpha = \frac{\tau}{I} = \frac{iBL^2/2}{mL^2/3} = \frac{3iB}{2m}$$

24. Ans (4)

At extreme position kinetic energy is zero, total energy in the form of potential energy.
At equilibrium kinetic energy is maximum but potential energy may or may not be zero.
(Example vertical oscillations of spring)

25. Ans (4)

$$U_i = \frac{q^2}{2 \times 2}, U_f = \frac{q^2}{2 \times (2+8)} = \frac{U_i}{5}$$

$$\% \text{ loss} = \frac{U_i - U_f}{U_i} \times 100 = \frac{U_i - U_i/5}{U_i} \times 100 = 80\%$$

26. Ans (4)

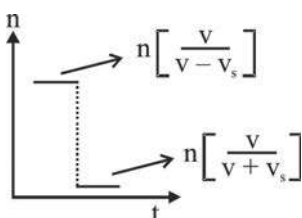
$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{(6\hat{i} + 6\hat{j} - 3\hat{k}) \cdot (7\hat{i} + 4\hat{j} + 4\hat{k})}{\sqrt{6^2 + 6^2 + 3^2} \sqrt{7^2 + 4^2 + 4^2}}$$

$$\cos \theta = \frac{42 + 24 - 12}{9 \times 9} = \frac{54}{9 \times 9} = \frac{2}{3}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \frac{\sqrt{5}}{3}$$

27. Ans (4)

$\xrightarrow{V_s}$ Observer Velocity = 0
Source
When Approaching = $n \left[\frac{V}{V - V_s} \right]$
When it's going away = $n \left[\frac{V}{V + V_s} \right]$



28. Ans (1)

$$1 \text{ kilo watt hour} = 10^3 \times 3600 \text{ Joule}$$

$$= 3.6 \times 10^6 \text{ joule}$$

29. Ans (4)

Brown.Black $\times$ 10 Green $\pm$ Gold
 $\Rightarrow 10 \times 10^5 \pm 5\% \Rightarrow 10^6 \pm 5\%$

30. Ans (3)

$$\Delta K = W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$$

$$= \int_{(2,3)}^{(3,0)} (3x^2\hat{i} + 4\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$= 3 \int_2^3 x^2 dx + 4 \int_3^0 dy = 7 \text{ Joule}$$

31. Ans (2)

$$\gamma_e = \gamma_{ph}$$

$$E_{ph} = \frac{hc}{\lambda_{ph}} \quad P_e = \frac{h}{\lambda_e}$$

$$\frac{E_{ph}}{P_e} = \frac{\frac{hc}{\lambda_{ph}}}{\frac{h}{\lambda_e}} = C$$

32. Ans (3)

$$I_1 = I_2 \Rightarrow \frac{2}{3} MR_1^2 = \frac{2}{5} MR_2^2$$

$$\Rightarrow 5R_1^2 = 3R_2^2 \Rightarrow \frac{R_1}{R_2} = \sqrt{\frac{3}{5}}$$

33. Ans (2)

$$(I_{res})_A = I + I + 2\sqrt{I}\sqrt{I}\cos 0^\circ$$

$$(I_{res})_A = 4I$$

$$(I_{res})_B = I + I + 2\sqrt{I}\sqrt{I}\cos 90^\circ$$

$$(I_{res})_B = 2I$$

$$\frac{(I_{res})_A}{(I_{res})_B} = \frac{4I}{2I} = \frac{2}{1}$$

34. Ans (2)

$$(1) a = \frac{50}{2} = 25 \text{ m/sec}^2$$

$$\xrightarrow{N} \boxed{B} \Rightarrow N = 1 \times a = 25 \text{ Newton}$$

$$(2) \boxed{\uparrow N} \quad N = 40 \text{ Newton}$$

$$(3) \boxed{\downarrow mg} \quad a = 25 \text{ m/sec}^2$$

$$N = m(ga+a) = 1(10+25)$$

$$= 35 \text{ Newton}$$

$$(4) \xrightarrow{N} \boxed{A}$$

$$N = ma$$

$$N = 1 \times 35 = 35 \text{ Newton}$$

35. Ans (3)

$$a + b = 14 \pm 0.14 \Rightarrow \% \text{ error} = \frac{0.14}{14} \times 100 = 1\%$$

$$a - b = 2 \pm 0.14 \Rightarrow \% \text{ error} = \frac{0.14}{2} \times 100 = 7\%$$

$$a \times b = 48 \pm 0.96 \Rightarrow \% \text{ error} = \frac{0.96}{48} \times 100 = 2\%$$

 so, order of % error $x < z < y$
36. Ans (3)

$$m\omega_{\max}^2 r = T_{\max}$$

37. Ans (1)

The direction of polarization is parallel to electric field

$$\therefore \vec{X} \parallel \vec{E}$$

 The direction of the propagation of the wave is parallel to $\vec{E} \times \vec{B}$ -

$$\therefore \vec{K} \parallel \vec{E} \times \vec{B}$$

38. Ans (1)

$$R = 500 \Omega$$

$$X_L = \omega \times L = 1000 \times 0.5 = 500$$

$$\cos \phi = \frac{R}{\sqrt{R^2 + X_L^2}}$$

39. Ans (4)

$$\bar{A} \cdot (\bar{A} + \bar{B}) = (\bar{A} + (\bar{A} + \bar{B})) = \bar{A} + \bar{B}$$

40. Ans (4)

For uniform temperature on heating the rod, K should be infinite.

41. Ans (1)

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P + \frac{1}{2}\rho v^2 = \frac{P}{2} + \frac{1}{2}\rho v'^2$$

$$\left(P - \frac{P}{2}\right) + \frac{1}{2}\rho v^2 = \frac{1}{2}\rho v'^2$$

$$v' = \sqrt{\frac{P}{\rho} + v^2}$$

42. Ans (3)

$$P_2 = P_1 \left(\frac{V_1}{V_2}\right)^\gamma$$

(Adiabatic because sudden process)

$$r = \frac{3}{2}, V_2 = \frac{V_1}{4}, P_1 = 1 \text{ atm}$$

$$P_2 = 1(4)^{3/2} = 8 \text{ atm}$$

43. Ans (2)

Potential energy = KE at large separation

$$\frac{1}{2} \times \frac{(e)(e)}{4\pi\epsilon_0 r} = \frac{\text{KE at large separation}}{2}$$

= KE of each proton

KE of each proton

$$= \frac{1}{2} \times \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{10^{-10}}$$

$$= 11.52 \times 10^{-19} \text{ J.}$$

44. Ans (4)

Magnetic field can be produced by moving charge and changing electric field.

45. Ans (4)

$$E = \frac{3NKT}{2}, V_{\text{rms}} = \sqrt{\frac{3KT}{m}}$$

$$T_1 = 300 \text{ K}, T_2 = 600 \text{ K}$$

$$\frac{E_2}{E_1} = \frac{T_2}{T_1} = 2 \Rightarrow E_2 = 2E_1$$

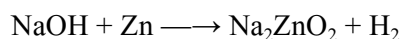
$$E_2 = 2 \times 6.21 \times 10^{-21} = 12.42 \times 10^{-21} \text{ J}$$

$$\frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{2} \Rightarrow V_2 = 484 \times \sqrt{2}$$

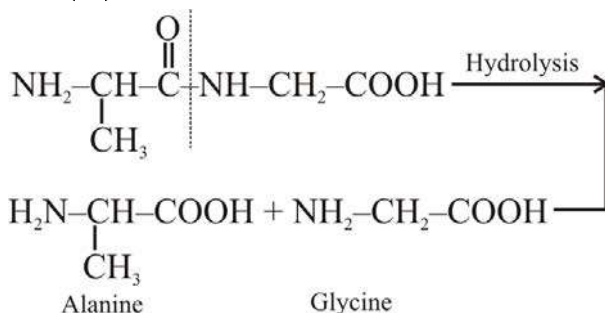
$$V_2 = 684 \text{ m/s}$$

47. Ans (1)

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$$

48. Ans (1)


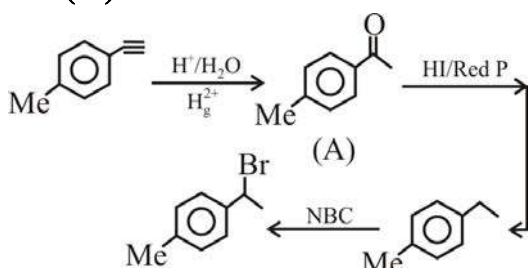
49. Ans (3)



50. Ans (3)

$$K_b = \frac{RTb^2}{1000L_v} \quad K_f = \frac{RT_f^2}{1000L_f}$$

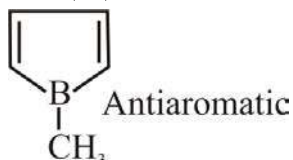
51. Ans (4)



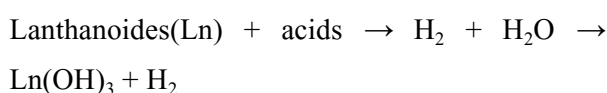
52. Ans (3)

Concept

53. Ans (2)



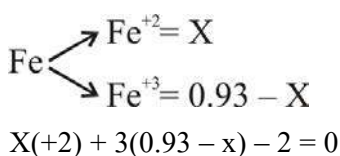
54. Ans (2)



55. Ans (1)

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56. Ans (1)

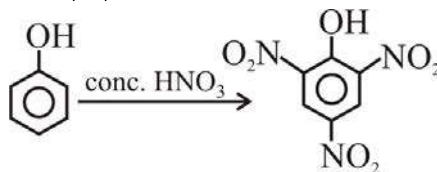


$$X = 0.79$$

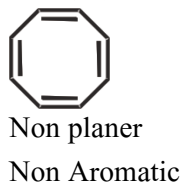
$$\text{Fe}^{+2} = 0.79 \quad \text{Fe}^{+3} = 0.14$$

$$\% \text{Fe}^{+3} = \frac{0.14}{0.93} \times 100 = 15\%$$

59. Ans (4)



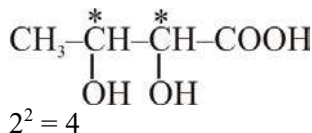
62. Ans (3)



63. Ans (4)

NCERT XIIth Pg. # 373

64. Ans (3)



66. Ans (3)

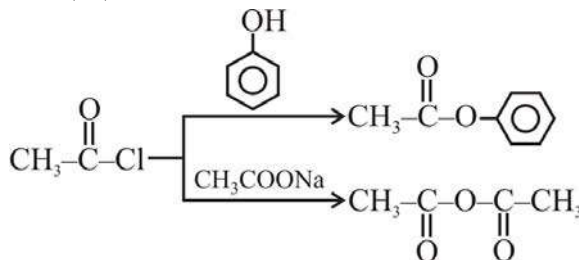
In CsCl $\text{Cs}^+ \rightarrow$ at body center

$\text{Cl}^- \rightarrow$ at every corner

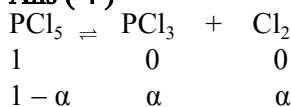
$$2(r_c + r_a) = \sqrt{3}a \text{ (along body diagonal)}$$

$$r_c + r_a = \frac{\sqrt{3}a}{2}$$

73. Ans (4)



75. Ans (4)



$$\sum n = 1 + \alpha$$

$$K_p = \frac{\alpha^2}{(1 - \alpha)} \times \left[\frac{P}{(1 + \alpha)} \right]^1$$

$$K_p = \frac{\alpha^2 P}{1 - \alpha^2} \simeq \alpha^2 P$$

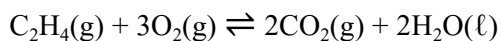
$$\alpha_1^2 P_1 = \alpha_2^2 P_2$$

$$(0.02)^2 \times 1 = (0.04)^2 P_2$$

$$\Rightarrow P_2 = 0.25 \text{ atm}$$

$$= \frac{1}{4} \text{ atm}$$

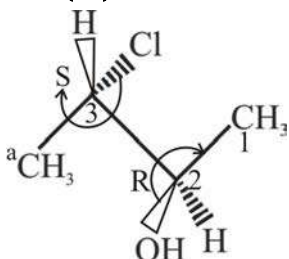
77. **Ans (2)**



$$\frac{560}{28} \text{ mol} \quad \frac{560}{28} \times 3 \text{ mol}$$

$$^m\text{O}_2 \frac{560 \times 3}{28} \times 32 = 1.92 \text{ kg}$$

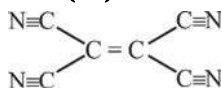
79. **Ans (3)**



80. **Ans (4)**

$$\frac{(t_{1/2})_I}{(t_{1/2})_{II}} = \left(\frac{[A_0]_{II}}{[A_0]_I} \right)^{n-1}$$

81. **Ans (3)**



No. of π bond = 9

No. of π electrons = 18

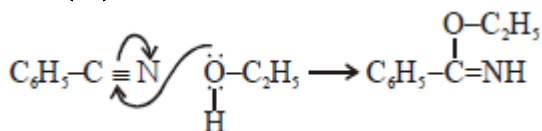
82. **Ans (2)**

In pseudo halide 'N' must be present.

85. **Ans (1)**

$$\Delta G = \Delta H - T.\Delta S$$

89. **Ans (3)**



90. **Ans (2)**

$\text{H}_2/\text{Ni}/\text{D}$ reduce - CHO and multiple bond.

94. **Ans (4)**

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97. **Ans (4)**

NCERT-XI- 317-318

104. **Ans (4)**

NCERT-XI- 101, 102, 104

105. **Ans (3)**

NCERT-XI- 292,293

107. **Ans (4)**

NCERT-XI- 104

118. **Ans (4)**

NCERT (XIth) Pg.#218-220

121. **Ans (4)**

NCERT-XI- 265, 264

130. **Ans (3)**

NCERT-XI- 271,272

133. **Ans (1)**

NCERT-XI- 314

135. **Ans (2)**

NCERT-XI- 314

138. **Ans (2)**

NCERT-XI- 324

139. **Ans (3)**

Module (E) Pg. # 138, (H) Pg. # 150

154. **Ans (3)**

NCERT-XI- 338

161. **Ans (3)**

NCERT-XI- 263

168. **Ans (3)**

NCERT-XI- 53, 54

170. **Ans (1)**

NCERT-XI- 331,332, 333

171. **Ans (2)**

NCERT-XI- 325

180. **Ans (3)**

NCERT-XI- 279